Network Traffic Analysis of Ransomware Activity in the Healthcare Sector Using Wireshark and MITRE ATT&CK

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Problem & Motivation

- Rising ransomware in healthcare
- Financial losses exceed $206 Million globally (2024)
- High-value sensitive data
- Need better detection methods

Objective of Study

- Identify ransomware behavior through network traffic analysis
- Use Wireshark (pcap analysis)
- Map findings to MITRE ATT&CK

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Dataset & Methodology

- Analyzed packet capture (.pcap) dataset
- Tools
  - Wireshark
  - MITRE ATT&CK framework
- Focus
  - Identify Indicators of Compromise (IoCs)

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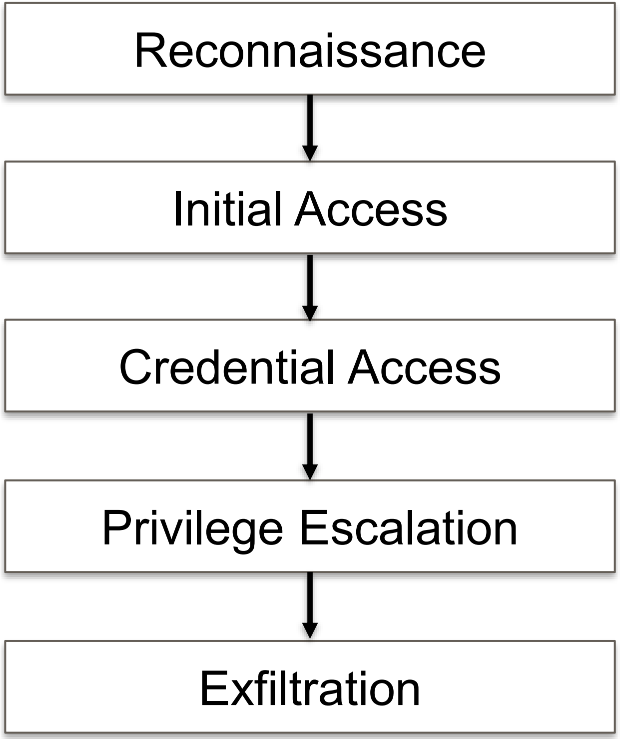
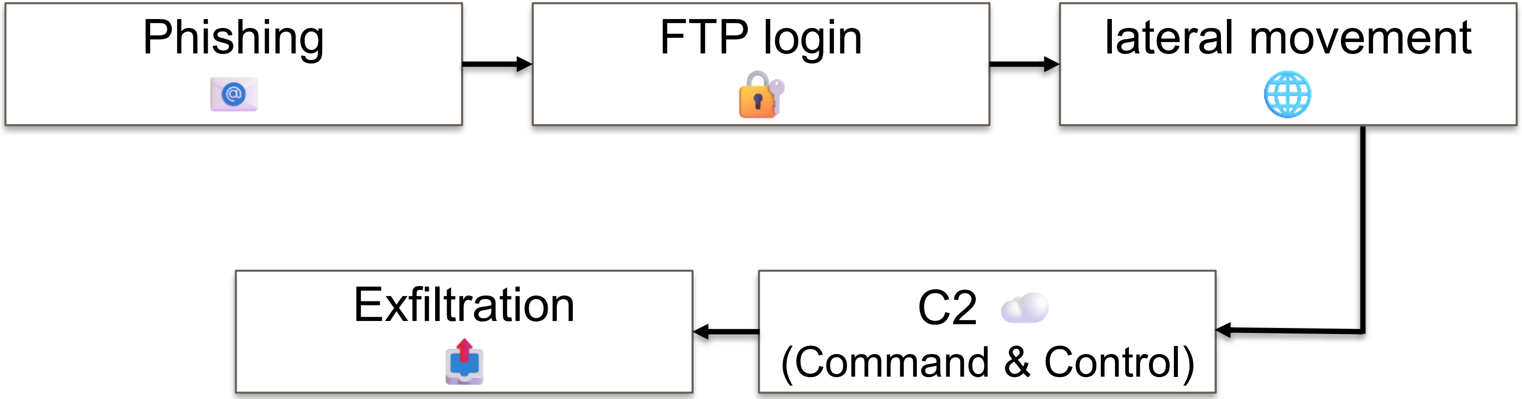
MITRE ATT&CK Overview



Attack Scenario



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KEY FINDINGS (Wireshark Evidence)

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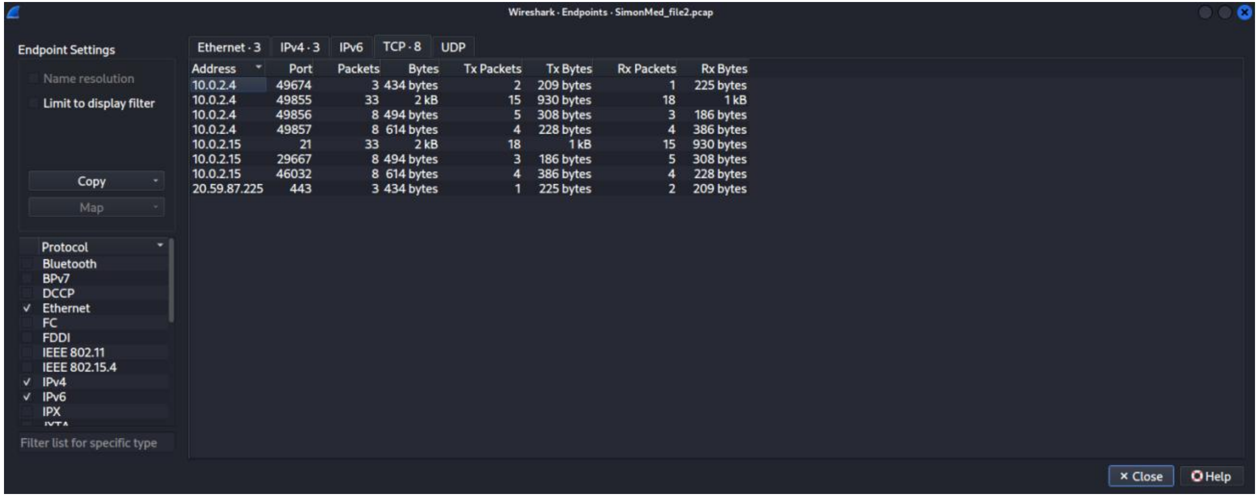
Suspicious Network Activity

- Internal host → Unfamiliar external IP (20.59.87.225)
- HTTPS traffic (possible C2)



Address	Port	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes
10.0.2.4	49674	3	434 bytes	2	209 bytes	1	225 bytes
10.0.2.4	49855	33	2 kB	15	930 bytes	18	1 kB
10.0.2.4	49856	8	494 bytes	5	308 bytes	3	186 bytes
10.0.2.4	49857	8	614 bytes	4	228 bytes	4	386 bytes
10.0.2.15	21	33	2 kB	18	1 kB	15	930 bytes
10.0.2.15	29667	8	494 bytes	3	186 bytes	5	308 bytes
10.0.2.15	46032	8	614 bytes	4	386 bytes	4	228 bytes
20.59.87.225	443	3	434 bytes	1	225 bytes	2	209 bytes

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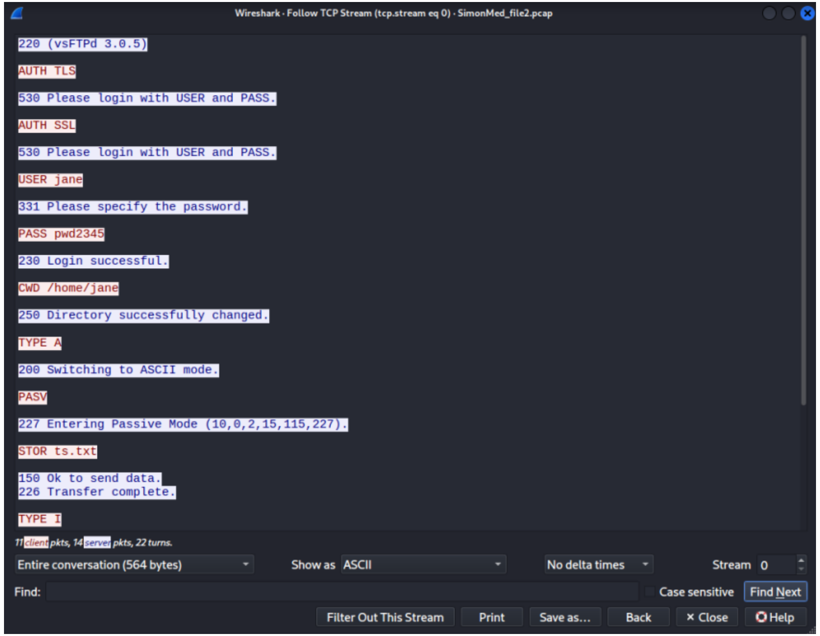
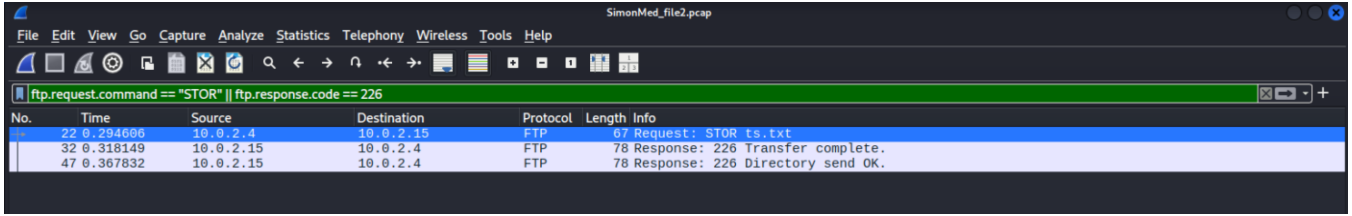
Credential Exposure

- FTP login in cleartext
  - USER: jane
  - PASS: pwd2345
- Confirmed login
  - 230 success



File Transfer / Exfiltration

- FTP STOR = Upload command
- Data transfer confirmed



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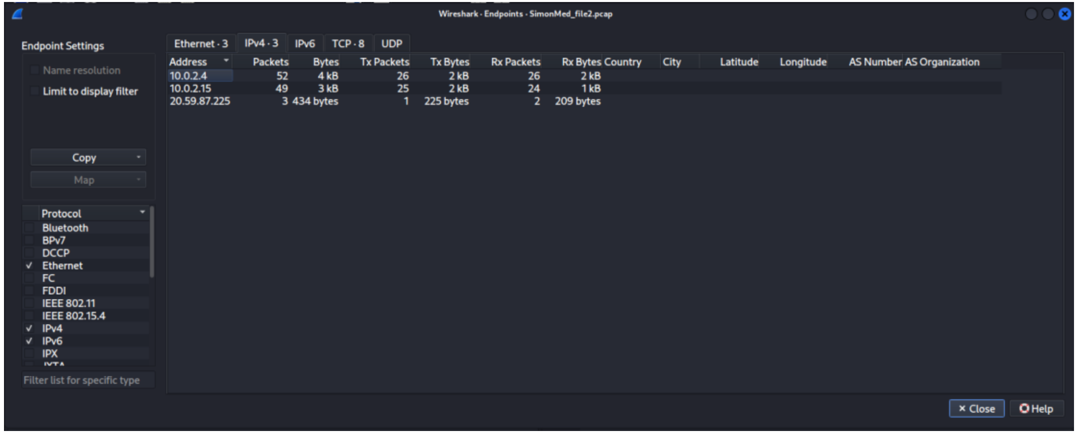
Lateral Movement

- Internal spread
  - 10.0.2.4 ↔ 10.0.2.15



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Summary Table

Severity	Finding	Evidence	Impact
High	FTP Authentication	*USER jane / PASS pwd2345* visible in TCP stream	Unauthorized access
High	FTP Login	*230 Login successful* response	Gained authenticated access
High	File Transfer	*STOR* command + *226 Transfer complete*	Data exfiltration
High	External Traffic	Connection to 20.59.87.225 via TCP 443	Possible external relay or C2 channel
Medium	Lateral Movement	10.0.2.4 accessing 10.0.2.15	Lateral movement and expand
Medium	Use of FTP	Unencrypted file transfer	Credential theft

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Key Takeaways

- Network traffic provides visibility into real attacker behavior
- MITRE ATT&CK helps structure analysis
- Weak protocols like FTP is a major risk

Conclusion

- Ransomware follows identifiable patterns
- Wireshark can expose critical IoCs
- MITRE ATT&CK improves structured analysis
- Healthcare networks need stronger monitoring

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Future Work

- Automate IoC detection
- Improve monitoring (especially insecure protocols)

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Q&A

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